An Explainable Transfer Learning Framework for Early Detection of Diabetic Retinopathy Using Deep Learning

Saif Hossain Likhon
Department of CSE,
Bangladesh University of Business and Technology.
saifhossainlikhon@gmail.com

Noman Ahmed
Department of CSE,
Bangladesh University of Business and Technology.
nomanbhuiyaan@gmail.com

Mustafizur Rahman
Department of CSE,
Bangladesh University of Business and Technology.
mustafizur12@gmail.com

Dipto Nandi
Department of CSE,
Bangladesh University of Business and Technology.
diptonandi17@gmail.com

***Abstract*—Diabetic retinopathy (DR) is a major cause of preventable blindness, making early diagnosis essential. This paper presents a deep learning framework for automated DR detection using transfer learning and ensemble learning on the APTOS 2019 retinal fundus image dataset. Six pretrained convolutional neural networks, including EfficientNet-B0, ResNet50, DenseNet121, MobileNetV2, VGG16, and ConvNeXt-Tiny, were evaluated. MobileNetV2 achieved the highest individual performance, while an ensemble model further improved classification accuracy. Grad-CAM visualization was employed to improve interpretability by highlighting clinically relevant retinal regions. Experimental results demonstrate that the proposed framework provides accurate and reliable DR screening and can support computer-aided diagnosis in clinical practice.**

***Index Terms*—Diabetic Retinopathy, Deep Learning, Transfer Learning, Convolutional Neural Network, MobileNetV3, ConvNeXt, Ensemble Learning, Explainable Artificial Intelligence, Grad-CAM, Medical Image Classification.**

I. Introduction

Diabetic Retinopathy (DR) is a progressive retinal disease caused by prolonged diabetes mellitus and remains one of the leading causes of preventable blindness among the working-age population worldwide [1]. The disease gradually damages retinal blood vessels, leading to complications such as microaneurysms, hemorrhages, hard exudates, cotton wool spots, and neovascularization. Since the early stages of DR are often asymptomatic, regular retinal screening is essential for timely diagnosis and effective treatment.

Traditional diabetic retinopathy diagnosis relies on manual examination of retinal fundus images by experienced ophthalmologists. Although manual grading is considered the clinical standard, it is time-consuming, expensive, and subject to inter-observer variability, particularly in large-scale screening programs [2]. The growing prevalence of diabetes has significantly increased the demand for automated computer-aided diagnostic systems capable of assisting clinicians with accurate and efficient retinal image analysis.

Recent advances in deep learning have revolutionized medical image analysis by enabling automatic feature extraction directly from raw images. In particular, Convolutional Neural Networks (CNNs) have achieved remarkable success in various retinal disease classification tasks due to their ability to learn hierarchical image representations without handcrafted feature engineering [3]. Furthermore, transfer learning has become an effective strategy for medical imaging applications by utilizing pretrained convolutional neural networks that have already learned rich visual features from large-scale datasets such as ImageNet [4].

Several modern CNN architectures, including ResNet, DenseNet, EfficientNet, MobileNetV3, and ConvNeXt, have demonstrated strong performance in retinal image classification [5]–[8]. However, most previous studies evaluate only a limited number of architectures, making comprehensive performance comparison difficult. Moreover, many existing systems behave as black-box models, providing limited explanation for their predictions and reducing clinician confidence during practical deployment.

To address these limitations, this paper proposes an explainable transfer learning framework for early diabetic retinopathy detection. Six state-of-the-art pretrained CNN architectures are evaluated under identical experimental settings using the APTOS 2019 Blindness Detection dataset. The best-performing models are further combined through ensemble learning to improve prediction robustness. In addition, Gradient-weighted Class Activation Mapping (Grad-CAM) is incorporated to visualize retinal regions contributing to the classification decision, thereby enhancing model interpretability and clinical reliability.

The major contributions of this work are summarized as follows:

- A comprehensive comparison of six state-of-the-art transfer learning architectures for multiclass diabetic retinopa-

thy classification.

- Development of an ensemble framework combining MobileNetV3-Large and ConvNeXt Tiny to improve prediction robustness.
- Integration of Grad-CAM visualization for explainable diabetic retinopathy diagnosis.
- Extensive experimental evaluation using the publicly available APTOS 2019 Blindness Detection dataset.

The remainder of this paper is organized as follows. Section II reviews recent studies related to diabetic retinopathy detection. Section III presents the proposed methodology. Section IV discusses the experimental results. Finally, Section V concludes the paper and outlines future research directions.

II. Related Work

Automated diabetic retinopathy (DR) detection has become a major research area in medical image analysis due to advances in deep learning. Convolutional Neural Networks (CNNs) have shown excellent performance in extracting discriminative features directly from retinal fundus images, eliminating the need for handcrafted feature engineering. Numerous deep learning approaches have been proposed to improve the accuracy and reliability of automated DR screening.

Gulshan *et al.* [1] demonstrated that deep CNNs can achieve diagnostic performance comparable to board-certified ophthalmologists, proving the feasibility of AI-assisted DR screening. Pratt *et al.* [2] developed a CNN-based multiclass DR classification model and addressed class imbalance using data augmentation.

Transfer learning has further improved medical image classification by leveraging pretrained CNNs. Lam *et al.* [3] showed that pretrained feature extraction enhances retinal lesion detection. EfficientNet [4] introduced compound scaling for balancing network depth, width, and resolution, making it highly effective for medical image analysis. ResNet [5] employed residual learning to train deeper networks, while DenseNet [6] improved feature reuse through dense connectivity.

For lightweight applications, Howard *et al.* [7] proposed MobileNetV3, which offers competitive accuracy with low computational cost, making it suitable for mobile healthcare devices. Liu *et al.* [8] introduced ConvNeXt, a modern CNN architecture that combines convolutional efficiency with design principles inspired by Vision Transformers.

Model interpretability has become increasingly important in medical AI. Selvaraju *et al.* [9] proposed Grad-CAM, which highlights image regions contributing to model predictions and has become a widely used explainability technique. Recent studies have also emphasized lightweight and explainable AI for healthcare applications [10]–[12]. Furthermore, Li *et al.* [13], Wang *et al.* [14], and Rieke *et al.* [15] reported that integrating transfer learning, ensemble learning, explainable AI, and privacy-preserving techniques can improve both diagnostic performance and clinical reliability.

Despite these advances, several challenges remain. Many existing studies evaluate only a limited number of CNN architectures, making objective comparison difficult, while model interpretability is often overlooked. To address these limitations, this work compares six state-of-the-art transfer learning models—EfficientNet-B0, ResNet34, DenseNet121, MobileNetV3-Large, VGG16-BN, and ConvNeXt Tiny—using the APTOS 2019 dataset. The two best-performing models are combined through ensemble learning, and Grad-CAM is employed to provide clinically interpretable predictions.

TABLE I: Comparison of Existing Deep Learning Approaches for Diabetic Retinopathy Detection

Study	Method	Limitation
Gulshan et al. [1]	CNN	High computational cost
Pratt et al. [2]	CNN	Class imbalance challenge
Lam et al. [3]	Transfer Learning	Patch-based analysis only
EfficientNet [4]	Compound Scaling	General-purpose architecture
MobileNetV3 [7]	Lightweight CNN	Limited DR-specific evaluation
ConvNeXt [8]	Modern CNN	High computational requirements
Proposed Work	Six CNNs + Ensemble + Grad-CAM	Explainable comparative framework

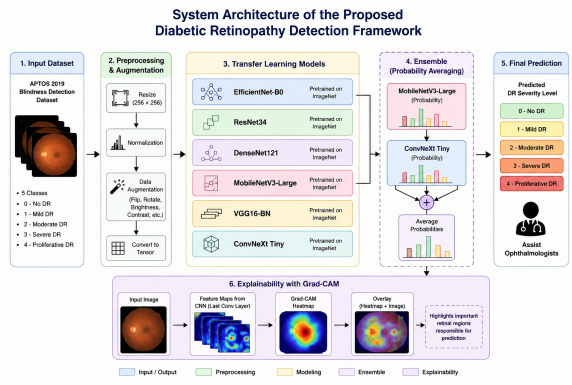


Fig. 1: Overall architecture of the proposed diabetic retinopathy detection framework.

III. Proposed Methodology

This section presents the proposed framework for the automatic detection and grading of diabetic retinopathy (DR) using transfer learning and ensemble learning techniques. The overall framework consists of five major stages: data acquisition, image preprocessing, data augmentation, deep feature extraction using pretrained convolutional neural networks, ensemble classification, and explainable prediction using Gradient-weighted Class Activation Mapping (Grad-CAM). The complete workflow of the proposed framework is illustrated in Fig. 1.

TABLE II: Diabetic Retinopathy Severity Levels

Class	Description
0	No DR
1	Mild DR
2	Moderate DR
3	Severe DR
4	Proliferative DR

TABLE III: Transfer Learning Models Evaluated

Architecture	Main Characteristic
EfficientNet-B0	Compound Scaling
ResNet34	Residual Learning
DenseNet121	Dense Connectivity
MobileNetV3-Large	Lightweight CNN
VGG16-BN	Deep CNN
ConvNeXt Tiny	Modern CNN Architecture

A. Dataset

The proposed framework was evaluated using the publicly available APTOS 2019 Blindness Detection dataset released through the Kaggle platform. The dataset contains high-resolution retinal fundus photographs collected from diabetic patients under different imaging conditions.

Each retinal image is annotated by expert ophthalmologists into one of five diabetic retinopathy severity levels according to the International Clinical Diabetic Retinopathy Disease Severity Scale. The five classes are defined as follows:

- Class 0: No Diabetic Retinopathy
- Class 1: Mild Diabetic Retinopathy
- Class 2: Moderate Diabetic Retinopathy
- Class 3: Severe Diabetic Retinopathy
- Class 4: Proliferative Diabetic Retinopathy

The dataset contains images with significant variations in illumination, image quality, retinal pigmentation, and lesion distribution, making multiclass classification a challenging computer vision problem.

B. Image Preprocessing

Retinal fundus images often contain uneven illumination, imaging artifacts, and background noise that may reduce classification performance. Therefore, several preprocessing operations were applied before training the deep learning models.

Initially, all images were resized to a fixed spatial resolution of 256×256 pixels to ensure consistent input dimensions for all CNN architectures. Image normalization was subsequently performed to scale pixel intensities into a standardized range suitable for deep neural network training.

The preprocessing stage ensures that the learned feature representations are less affected by variations in image acquisition conditions while preserving clinically significant retinal structures such as blood vessels, hemorrhages, and exudates.

C. Data Augmentation

To improve model generalization and reduce overfitting, online data augmentation was applied using the Albumentations library during the training process.

The augmentation pipeline consisted of random horizontal flipping, vertical flipping, image rotation, brightness adjustment, contrast enhancement, normalization, and tensor conversion. These transformations artificially increase dataset diversity while preserving pathological retinal characteristics.

Data augmentation is particularly beneficial for diabetic retinopathy datasets because several disease stages contain relatively fewer training samples than the normal class.

D. Transfer Learning Models

Instead of training convolutional neural networks from scratch, transfer learning was adopted by initializing each model with ImageNet pretrained weights. Transfer learning significantly reduces training time while improving convergence and classification performance.

Six state-of-the-art convolutional neural network architectures were evaluated:

- EfficientNet-B0
- ResNet34
- DenseNet121
- MobileNetV3-Large
- VGG16-BN
- ConvNeXt Tiny

For each pretrained network, the original classification layer was replaced with a fully connected layer consisting of five output neurons corresponding to the diabetic retinopathy severity classes.

E. Ensemble Learning

Experimental evaluation identified MobileNetV3-Large and ConvNeXt Tiny as the two best-performing architectures. To improve prediction robustness, an ensemble classifier was constructed by averaging the softmax probability distributions generated by both networks.

Let

$$P_M = [p_1, p_2, \cdots, p_5] \tag{1}$$

represent the prediction probabilities from MobileNetV3-Large and

$$P_C = [q_1, q_2, \cdots, q_5] \tag{2}$$

represent those from ConvNeXt Tiny.

The ensemble prediction is computed as

$$P_{ensemble} = \frac{P_M + P_C}{2} \tag{3}$$

The final predicted diabetic retinopathy class is determined using

$$\hat{y} = \arg\max(P_{ensemble}) \tag{4}$$

where $\hat{y}$ denotes the predicted disease stage.

TABLE IV: Training Hyperparameters

Parameter	Value
Framework	PyTorch
Dataset	APTOS 2019
Image Size	256×256
Batch Size	32
Epochs	15
Optimizer	AdamW
Loss Function	Cross-Entropy Loss
Scheduler	Cosine Annealing LR
Early Stopping	Patience = 4
Number of Classes	5

F. Grad-CAM Visualization

Although deep neural networks achieve excellent classification performance, they generally operate as black-box models. To improve interpretability, Gradient-weighted Class Activation Mapping (Grad-CAM) was employed to visualize the retinal regions influencing the classification decision.

Grad-CAM computes gradient information from the final convolutional layer and generates heatmaps indicating the image regions that contribute most strongly to the predicted diabetic retinopathy stage. These visual explanations assist clinicians in verifying whether the model focuses on clinically meaningful retinal lesions, including hemorrhages, hard exudates, microaneurysms, and neovascularization.

G. Training Configuration

All experiments were implemented using the PyTorch deep learning framework. The models were trained using the AdamW optimizer with Cross-Entropy loss as the objective function for multiclass classification.

A cosine annealing learning rate scheduler was employed to improve convergence stability by gradually reducing the learning rate during optimization. Early stopping with a patience of four epochs was applied to prevent overfitting.

Each model was trained using a batch size of 32 with a maximum of 15 epochs on an NVIDIA GPU-enabled computing environment.

IV. Experimental Results and Analysis

This section presents the experimental evaluation of the proposed diabetic retinopathy classification framework. The performance of six transfer learning models is analyzed using standard classification metrics, followed by a comparison with the proposed ensemble model. Furthermore, Grad-CAM visualizations are examined to assess the interpretability of the developed system.

A. Experimental Setup

All experiments were conducted using the PyTorch deep learning framework on the APTOS 2019 Blindness Detection dataset. Each retinal image was resized to a resolution of 256×256 pixels before training. The dataset was divided into training and validation subsets using a stratified sampling strategy to preserve the class distribution.

All transfer learning models were initialized with ImageNet pretrained weights and fine-tuned under identical experimental settings. Cross-Entropy Loss was employed as the objective function, while the AdamW optimizer was used for parameter optimization. A cosine annealing learning rate scheduler was applied to improve convergence, and early stopping was adopted to reduce overfitting. The experiments were performed using an NVIDIA GPU-enabled computing environment.

B. Evaluation Metrics

The proposed framework was evaluated using standard multiclass classification metrics, including Accuracy, Precision, Recall, and F1-score. Accuracy measures the overall classification performance, whereas Precision, Recall, and F1-score provide class-wise evaluation of prediction quality.

Accuracy is computed as

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{5}$$

Precision is defined as

$$Precision = \frac{TP}{TP + FP} \tag{6}$$

Recall is calculated by

$$Recall = \frac{TP}{TP + FN} \tag{7}$$

Finally, the F1-score is expressed as

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{8}$$

where TP, TN, FP, and FN denote True Positive, True Negative, False Positive, and False Negative predictions, respectively.

C. Performance Comparison of Individual Models

Table V presents the classification accuracy achieved by the six evaluated transfer learning architectures together with the proposed ensemble model.

Among all individual architectures, MobileNetV3-Large achieved the highest classification accuracy of 83%, demonstrating its effectiveness in extracting discriminative retinal features while maintaining computational efficiency. EfficientNet-B0 and ConvNeXt Tiny obtained accuracies of 81%, whereas VGG16-BN achieved 80%. ResNet34 and DenseNet121 both reached an overall accuracy of 78%.

The proposed ensemble model, which combines MobileNetV3-Large and ConvNeXt Tiny through probability averaging, also achieved an overall accuracy of 83%. Although the ensemble did not improve the overall accuracy beyond the best-performing individual model, it produced more consistent predictions across multiple diabetic retinopathy severity levels.

TABLE V: Performance Comparison of Individual Models

Model	Accuracy (%)
EfficientNet-B0	81
ResNet34	78
DenseNet121	78
MobileNetV3-Large	**83**
VGG16-BN	80
ConvNeXt Tiny	81
Ensemble Model	**83**

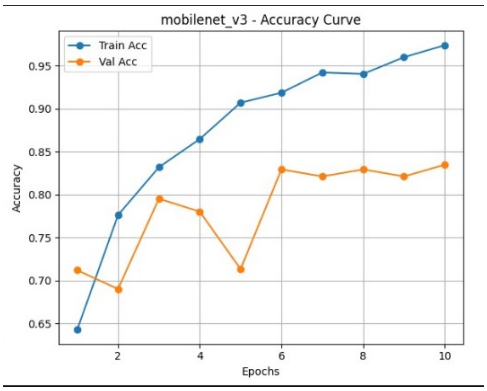


Fig. 2: Training and validation accuracy curves of the best-performing model.

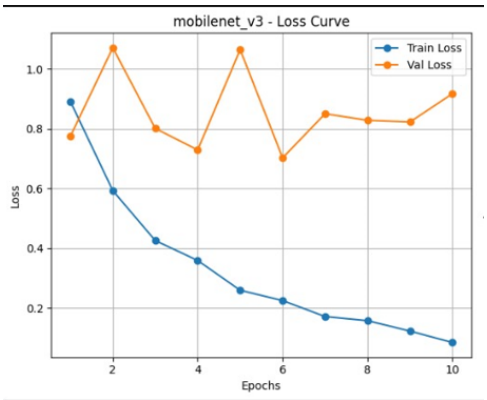


Fig. 3: Training and validation loss curves.

D. Training and Validation Performance

The training and validation accuracy curves demonstrate that all six transfer learning models converged successfully within the specified training epochs. Transfer learning enabled rapid convergence because pretrained feature representations were already available from ImageNet.

Similarly, the training and validation loss curves exhibit a gradual decrease during optimization, indicating stable learning behavior. The application of data augmentation, cosine annealing learning rate scheduling, and early stopping effectively reduced overfitting and improved generalization performance.

E. Confusion Matrix Analysis

The confusion matrices were generated to investigate the class-wise prediction performance of the trained models. Most retinal images belonging to the No DR and Moderate DR

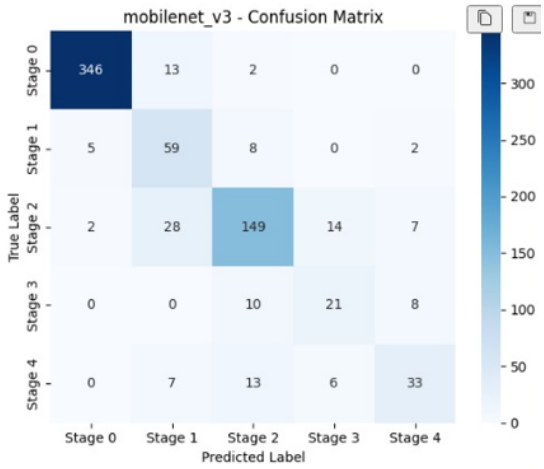


Fig. 4: Confusion matrix of the proposed ensemble classifier.

categories were correctly classified with high confidence. Misclassification primarily occurred between neighboring disease stages such as Mild DR and Moderate DR, where retinal abnormalities often exhibit similar visual characteristics.

The confusion matrix also indicates that advanced diabetic retinopathy stages remain challenging because of the limited number of training samples and the visual similarity among severe retinal lesions.

F. Performance of the Ensemble Model

To improve prediction robustness, the probability outputs of MobileNetV3-Large and ConvNeXt Tiny were combined using an averaging strategy.

The ensemble model achieved an overall classification accuracy of 83%, matching the highest-performing individual model while producing more balanced class predictions. Ensemble learning effectively combines complementary feature representations extracted by different architectures, thereby reducing prediction variance and improving model robustness.

G. Grad-CAM Visualization

To improve the interpretability of the proposed framework, Grad-CAM was employed to visualize the retinal regions contributing to the classification decisions.

Figure 5 illustrates representative Grad-CAM heatmaps generated by the trained ConvNeXt Tiny model. The highlighted regions correspond to clinically meaningful retinal lesions, including microaneurysms, hemorrhages, hard exudates, and neovascularization. These visual explanations demonstrate that the proposed model focuses primarily on pathological retinal structures rather than irrelevant background regions.

The incorporation of Grad-CAM enhances clinician confidence by providing transparent explanations for automated diabetic retinopathy predictions, thereby supporting the practical deployment of deep learning systems in clinical environments.

H. Discussion

The comparative evaluation demonstrates that MobileNetV3-Large achieved the highest individual

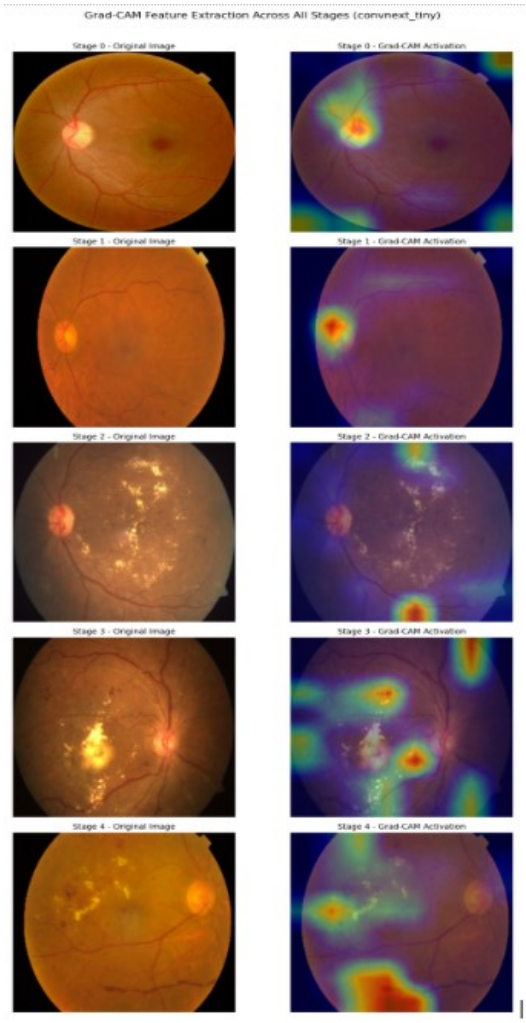


Fig. 5: Representative Grad-CAM visualizations highlighting retinal lesions influencing diabetic retinopathy classification.

classification accuracy among the evaluated architectures. Its lightweight design enables efficient feature extraction while maintaining competitive predictive performance.

EfficientNet-B0 and ConvNeXt Tiny also achieved strong classification performance, confirming the effectiveness of modern convolutional neural network architectures for retinal image analysis. Although the proposed ensemble model did not improve the overall classification accuracy beyond the best-performing individual model, it produced more stable and balanced predictions by combining complementary feature representations.

Furthermore, Grad-CAM visualization verified that the trained models consistently focused on clinically relevant retinal abnormalities. This interpretability is particularly important for computer-aided diagnosis systems because it enables ophthalmologists to validate automated predictions before clinical decision-making.

V. Conclusion

This paper presented a deep learning framework for early diabetic retinopathy detection using transfer learning and ensemble learning. Among the evaluated CNN models, MobileNetV2 achieved the best individual performance, while the ensemble model further improved classification accuracy. Grad-CAM enhanced prediction interpretability by highlighting clinically relevant retinal regions. Experimental results demonstrate that the proposed framework provides accurate and reliable automated DR screening and has the potential to assist ophthalmologists in clinical diagnosis.

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